

Jeffrey Wang

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EDUCATION

Carnegie Mellon University

Expected May 2027

Bachelor of Computer Science (3.69/ 4.0)

Pittsburgh, PA

- **Computer Graphics**, Computer Vision, **Computer Systems**, Parallel and Sequential **Data Structures and Algorithms**

EXPERIENCES

Software Engineer Intern

May 2025 – Nov. 2025

Center of Transformational Play

Pittsburgh, PA

- Worked with a cross-functional team of 12 to build an **offline-capable iOS/Android prototype** in **C#** with **~10 hours of gameplay**, optimized to run smoothly on **5-6-year-old devices**,
- Optimized builds to **maintain 60 FPS** on iPhone 8 / Ipad Minis/Galaxy Tab s8/ Pixel 7
- Reduced content integration time **90% (10 min to 1 min)** by building a **Python CSV import and validation pipeline** for dialogue, triggers, events, and item attributes.

Software Engineer Intern

Oct. 2024 – May 2025

Carnegie Mellon University Software Engineering Institute

Pittsburgh, PA

- Rapidly prototyped a **modular interactive assessment** (Unity) for the **DoD** to evaluate **AI and data literacy competencies**.
- Collaborated with stakeholders in an **Agile sprint cadence** to refine requirements, prioritize scope, and deliver iterative builds aligned with KSAT practices.

PROJECTS

Custom Real-time Render and Physics Engine

Oct. – Dec 2025

- Implemented a real-time rendering pipeline in **C++/OpenGL** with custom **GLSL shaders** for forward lighting, dynamic shadow mapping, and framebuffer post-processing.
- Profiled and optimized a soft-body + metaball fragment-shader ray-marching pipeline, **improving frame rate by 30%** at native display resolution for **macOS, windows, and linux** builds.
- Built a CI pipeline with **GitHub Actions** to produce builds for (macOS/Windows/Linux),

O-Quest — Geolocation Orientation Scavenger Hunt (Web-Native)

May - Aug. 2025

- Built and maintained a **Rust + Postgres** backend for a geolocation scavenger hunt, implementing a modular **handlers/services** architecture and operational tooling (**CSV-based challenge import; QR export pipeline**) to support rapid iteration and event setup.
- Optimized read-heavy endpoints (challenge feed, progress, leaderboard) with query shaping and indexing to maintain low latency with **600 concurrent users**

Scottylab CMUMaps (interactive campus navigation application)

Oct. 2024 – Aug. 2025

- Built indoor/outdoor navigation features and authentication workflows using **Node.js** and **MongoDB**, supporting production usage.
- Implemented UI/UX redesign with **React** from Figma specs; increased daily active users to **400**.

SKILLS

Programming Languages: Python, C/ C++, C#, JavaScript/Typescript , **HTML5, CSS**

Skills: SQL, **Git**, Matplotlib, Plotly | **Frameworks:** React, Node.js